

Introductory concepts

1

An alphabet is: 
(1 Ponto)

- ☐ An infinite set of symbols
- ☐ A finite set of symbols

2

The set of symbols that can be used in a programming language is an alphabet 
(1 Ponto)

- ☐ TRUE
- ☐ FALSE

3

A string is: 
(1 Ponto)

- ☐ an infinite sequence of symbols
- ☐ a finite sequence of symbols

4

Given the string mieic 
(1 Ponto)

- ☐ $|mieic| = 5$
- ☐ $|mieic| = 6$

5

A language is: 
(1 Ponto)

- ☐ A finite set of strings over an alphabet
- ☐ A set (finite or infinite) of strings over an alphabet

6

The set $\{a, b, c, ab, bc\}$ is: 
(1 Ponto)

- ☐ A language over the alphabet $\{a,b,c\}$
- ☐ An alphabet

7

Given $S = \{a, b\}$, the Kleen-star closure (*) on S , i.e., S^* is (the symbol 'e' represents the epsilon, empty string) 
(1 Ponto)

- ☐ $\{e, a, b, ab, ba\}$
- ☐ all the strings over the alphabet $\{a,b\}$
- ☐ $\{a, b, aa, bb, ab, ba\}$
- ☐ all the strings over the alphabet $\{a,b,c\}$

8

Given an alphabet, any language over that alphabet:
(1 Ponto)

- ☐ consists of the strings given by the kleene-star closure on the alphabet
- ☐ is a subset of the strings given by the kleene-star closure on the alphabet

9

The language given by: $\{w \mid |w| = 2 \text{ and } w \in \{a,b\}^*\}$
(1 Ponto)

$\{w \mid |w| = 2 \text{ and } w \in \{a,b\}^*\}$

(equivalent to $\{w \mid |w| = 2, w \in \{a,b\}^*\}$ and to $\{w \in \{a,b\}^* \mid |w| = 2\}$)

- ☐ is represented by the set {aa, bb, ab, ba}
- ☐ is represented by the set {ab, ba}

10

Given the language $L =$
(1 Ponto)

$\{w \mid w \text{ consists of an odd number of 0s}\}$

- ☐ the set of strings {0, 00, 000}
- ☐ the sets of strings {0, 000, 000000} belongs to L